



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026-27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 12

Team No.: 15

Project Title: Linux System Information and Resource Monitoring Tool

Team Members

Roll Number	Student Name	Signature
2520090216	J Pragathi	J. Pragathi
2520090214	N Charishma	N. Charishma
2520090212	J Rakshitha	J. Rakshitha

1. Abstract

The Linux System Information and Resource Monitoring Tool is a Linux-based system programming project designed to monitor and display important system resources and operating system information through a command-line interface. Managing and understanding system resources such as CPU, memory, processes, storage, and system activity is essential for efficient system operation and performance analysis.

The proposed tool will collect and display real-time information about CPU utilization, memory usage, running processes, disk and file-system information, and other system-level resources. The project will demonstrate important Operating Systems and Systems Programming concepts including system calls, process management, inter-process communication, signals, virtual memory, file I/O, POSIX threads, synchronization, mutexes, and semaphores.

The system will be implemented in C on Linux/Ubuntu, using appropriate Linux system calls and POSIX APIs. A modular design will be followed so that different monitoring functions can operate independently while demonstrating the relevant OS concepts.

The expected outcome is a functional command-line monitoring tool that provides useful system information while demonstrating practical implementation of Linux operating-system mechanisms. The project will also provide hands-on understanding of processes, memory, file systems, communication, and concurrency in a Linux environment.



2. Problem Statement

Linux systems manage several resources such as CPU, memory, processes, storage, and file systems continuously. However, understanding the current state and usage of these resources through basic commands can be difficult for users who need a simple and organized view of system activity.

The proposed project addresses this problem by developing a Linux-based System Information and Resource Monitoring Tool that collects and displays important system and resource information through a command-line interface. The project will demonstrate how Operating Systems provide access to system resources through system calls, processes, memory management, file I/O, inter-process communication, and POSIX mechanisms.

The project aims to provide a practical way to observe system resources while applying core Operating Systems and Systems Programming concepts in a Linux environment.

3. Objectives

1. To develop a Linux-based command-line tool for collecting and displaying system information and resource usage.
2. To monitor CPU, memory, processes, storage, and file-system resources using Linux system calls and APIs.
3. To demonstrate Operating Systems concepts such as process management, inter-process communication, signals, virtual memory, file I/O, threads, and synchronization.
4. To provide a modular and user-friendly monitoring tool that demonstrates practical Linux/POSIX systems programming concepts.

4. Proposed Methodology

The project will be implemented as a modular command-line application in C on the Linux/Ubuntu platform. The system will collect information from the Linux operating system using appropriate system calls, POSIX APIs, and system interfaces.

The application will provide a menu-driven interface through which the user can select different monitoring functions such as system information, CPU usage, memory usage, process information, disk and file-system information, and other system resources.

The project will be divided into modules for process monitoring, memory monitoring, file and disk monitoring, inter-process communication, signal handling, and thread-based monitoring. POSIX threads and synchronization mechanisms will be used where required to demonstrate concurrent monitoring. Pipes/FIFOs and signals will be incorporated to demonstrate communication and control between processes.

The collected information will be processed and displayed in a clear command-line format. Each module will be tested independently, followed by integration testing of the complete

system. The final application will demonstrate practical Operating Systems and Linux systems programming concepts through a working monitoring tool.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
System Calls	To interact with the Linux operating system and obtain system-level information.
Process Management	To monitor running processes, process IDs, parent processes, and process states.
Fork () / exec ()	To demonstrate process creation and execution.
Pipes / FIFOs	To demonstrate inter-process communication between monitoring processes.
Signals	To provide process control and asynchronous notifications.
Virtual Memory	To monitor and display memory and virtual memory usage.
File I/O System Calls	File I/O System Calls
File Descriptors	To demonstrate Linux file and I/O management.
POSIX Threads (pthread)	To perform different monitoring operations concurrently.
Mutexes	To protect shared data accessed by multiple threads.
Semaphores	To demonstrate synchronization between concurrent operations.
Process Synchronization	To coordinate multiple processes and monitoring activities.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520090216	J Pragathi	System and Process Monitoring
2520090214	N Charishma	Memory and File-System Monitoring
2520090212	J Rakshitha	IPC and Multithreading

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system environment for developing and running the project
C / C++	Implementing the monitoring tool and OS-level functionality.

GCC Compiler	Compiling the C source code
POSIX APIs	Implementing processes, threads, IPC and synchronization.
Makefile	Building and managing the project source code
Git / GitHub	Version control and project repository

8. Expected Outcome

The project is expected to provide a functional Linux-based command-line tool for monitoring and displaying important system information and resource usage. The tool will provide information about CPU, memory, running processes, file systems, disk usage, and other system resources. It will also demonstrate IPC, signal handling, multithreading, and synchronization using Linux/POSIX mechanisms. The expected outcome is a working monitoring system that provides clear resource information while demonstrating practical Operating Systems and Systems Programming concepts in a Linux environment.

9. GitHub Repository

GitHub Repository Link:

<https://github.com/charishma18-n/KLH-CSIT-2026-15-SystemResourceMonitor>

Repository Created: Yes

☒ Yes ☐ No

Faculty Name and Signature

Faculty Name: M. Raghupathi

Signature: [Signature]

Remarks:

Faculty Use

Project Approved: Approved

☒ Yes ☐ No

Date: 2/9/2016

Faculty Signature: [Signature]